

Quantum Meta-Reinforcement Learning via Interference-Driven Policy Architectures

Chetan Guduru

Most reinforcement learning systems learn by trial and error: they update policies through repeated training until they perform well on a specific task. This often leads to brittle behavior; when the environment changes, performance collapses. This project asks a different question: can a system generalize across tasks without training at all?

Inspired by physics, I explore whether quantum interference, the phenomenon that allows waves to combine constructively or destructively, can act as a built-in mechanism for generalization in reinforcement learning.

I design a fixed-policy quantum reinforcement learning framework in which the agent’s behavior is determined entirely by the structure of a small quantum circuit. There are no gradients, learning updates, or parameter tuning. Instead, different actions interfere with one another, shaping behavior through quantum phase relationships rather than trial-and-error learning.

To evaluate this idea, I test six quantum policy circuits and compare them to classical baselines (greedy search, random choice, and Q-learning) across three grid-world environments:

1. a static goal,
2. a moving target,
3. a sparse environment with penalties.

While classical agents perform well only when rewards are stable or dense, the quantum interference-based agents remain stable in environments that change or penalize rigid strategies. Notably, interference-structured policies generalize better to new tasks even though they were never trained on them.

To explain why this happens, I develop a minimal two-qubit Hamiltonian model linking quantum physics to learning behavior. I show analytically that generalization is governed by the curvature of the agent’s quantum state, quantified by the relation:

$$\text{Tr}(F_Q) = 4 \text{Var}(H)$$

This relation connects quantum Fisher information, a measure of sensitivity to change, to Hamiltonian variance. Empirically, this curvature predicts whether an agent adapts smoothly across tasks or over-specializes.

These results suggest that quantum interference itself can serve as an inductive bias, enabling adaptation without learning. More broadly, this work proposes a physics-grounded perspective on generalization with implications for quantum machine learning, meta-learning, and interpretable AI.